

② Shared data :

```
bool waiting[2] = {false};
int turn = 0;
```

P0:

```
while (true)
```

```
{
  a1 waiting[0] = true;
```

```
  a2 while (turn != 0)
```

```
  {
    a3 while (waiting[1]);
```

```
    a4 turn = 0; ←
```

```
  }
```

```
  // CS
```

```
  waiting[0] = false;
```

```
  // RS
```

```
}
```

P1:

```
while (true)
```

```
{
  b1 waiting[1] = true;
```

```
  b2 while (turn != 1)
```

```
  {
    b3 while (waiting[0]);
```

```
    b4 turn = 1;
```

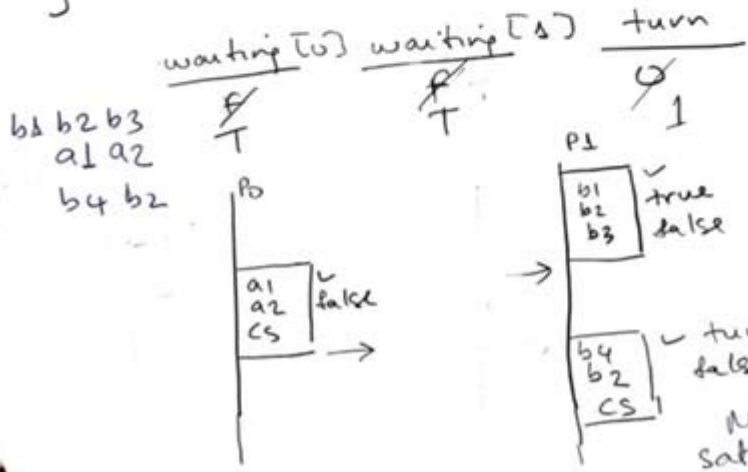
```
  }
```

```
  // CS
```

```
  waiting[1] = false
```

```
  // RS
```

```
}
```



①

Shared data:

ABCABCABC...

Semaphore $S_1 = 0, S_2 = 0, S_3 = 1$

Process A:
do {
 wait(S_3);
 // A
 signal(S_1);
} while (true);

Process B:
do {
 wait(S_1);
 // B
 signal(S_2);
} while (true);

Process C:
do {
 wait(S_2);
 // C
 signal(S_3);
} while (true);